Ahmed Rahi

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PERSONAL BIO

Software Engineer at Quantium with an MSc in Advanced Computer Science (Artificial Intelligence) from the University of Leeds and a First-Class BSc in Computer Science from the University of Leicester. Experienced in full-stack development across the entire software lifecycle, having delivered features and led architectural migrations on a retail analytics platform serving thousands of clients. Passionate about building scalable, high-quality software and exploring emerging technologies.

EDUCATION

University of Leeds Leeds
Master of Science Sep 2023 – Aug 2024
Advanced Computer Science (Artificial Intelligence)
Relevant Modules: Algorithms; Data Mining and Text Analytics; Data Science; Deep Learning; Machine Learning; Blockchain Technologies; Cloud Computing Systems; Knowledge Representation and Reasoning.

University of Leicester Leicester
Bachelor of Science Sep 2020 – Jul 2023
Computer Science
First-Class Honours
Relevant Modules: C++ Programming; Internet and Cloud Computing; Algorithms, Data Structures, and Advanced Programming; Operating Systems and Networking; Databases and Domain Modelling.

EMPLOYMENT

Quantium Manchester

Software Engineer (Level 2) Sep 2026 – Present

- Architected and built 'Proactive Insights' from scratch in C# (.NET 10): an event-driven backend module using RabbitMQ to asynchronously process post-hydration data and send real-time analytical alerts to users, supporting every alert in the platform's metrics library.
- Led a large-scale ASDA database migration covering millions of records, moving from legacy Walmart product IDs to custom IDs to decouple and standardise product management across the platform.
- Redesigned Fast Reporting end-to-end with a sleek, modern UI and on-demand loading, fetching only the report selected in the header rather than every report up front, drastically reducing Snowflake query costs.
- Built custom Claude Code skills and automated workflows for extending the backend, cutting the time to add a new alert type from 3–4 days to 1 day.

Software Engineer (Level 1) Sep 2025 – Sep 2026

- Led the migration of checkout from a monolithic frontend to a Micro-Frontend (MFE) architecture using Single-SPA, enabling multiple teams to develop and deploy their parts of the platform independently.
- Built and maintained frontend components in React using an in-house design system, keeping the user experience consistent across the platform.
- Maintained and cleaned up CI/CD pipelines in GitHub Actions, simplifying build and deployment workflows across backend and frontend services.
- Used Datadog for application monitoring and observability, giving the team visibility into performance and errors across frontend and backend services.
- Wrote and maintained end-to-end test suites in Cypress, ensuring reliability across frontend modules.
- Developed backend services in C#, contributing to full-stack feature delivery.

Graduate Software Engineer Sep 2024 – Sep 2025

- Delivered a full redesign of the Watchlist feature within checkout, a retail product analytics platform used by thousands of clients, improving usability and visual consistency throughout the application.
- Modernised the Dashboard UI to be more scalable and interactive, adding drag-and-drop layout customisation using TanStack.
- Supported the migration of Dashboard backend services to .NET 8, alongside completing the graduate programme's training in Git, automation, testing, security, DevOps, cloud and SQL.
- Worked in an Agile team, taking part in sprints, stand-ups and code reviews to deliver iterative improvements.

PROJECTS

Kinect Fitness Playground Jan 2026

- Built a fitness application in Python using the Microsoft Xbox Kinect V2 sensor for real-time body movement tracking, supporting exercises including squats, jumping jacks and bicep curls with automatic rep counting and form feedback.
- Implemented experimental heart rate monitoring using Photoplethysmography (PPG) via visual detection through the Kinect camera.
- Developed a personal leaderboard and performance tracking system that stores session history, rep counts and form quality scores.
- Used OpenCV, NumPy and SciPy for computer vision and signal processing, and Pygame for the GUI.
- Designed a modular, extensible architecture so new exercises can be added easily, making it straightforward for open-source contributors to extend.

Edustruct (IC Hack 2024 Project) Nov 2023

- Built at IC Hack 2024: an educational platform that teaches children to code in a visual, interactive way.
- The project merges the principles of Leetcode with the simplicity of Scratch, offering a more complete and traditional coding language experience.
- Designed to provide short 5-minute exercises daily, engaging children whilst teaching them coding concepts.
- Implemented with JavaScript, Flask and HTML, with work split across frontend, backend and testing sub-teams.

2D Game Engine (Covert Woods) Nov 2022

- Designed and built a 2D game engine in C# and MonoGame with physics and collision handling, and used it to create the game 'Covert Woods'.
- Completed as my undergraduate dissertation.

ADDITIONAL

Technical Skills:
Languages: C#, TypeScript, JavaScript, Python, SQL, Swift
Frameworks: .NET 10, React, Single-SPA, TanStack, SwiftUI, Flask
Cloud & tools: Azure, AWS, Docker, RabbitMQ, GitHub Actions, Datadog, Cypress, Snowflake, MySQL, SQLite, Claude Code
Practices: REST APIs, event-driven architecture, micro-frontends, CI/CD, Agile/Scrum

Technical Blogging – Medium (@ahmedrahi)
Authored technical articles on Medium covering hardware architecture and personal projects.

- Kinect Fitness Playground: Creating an Interactive Exercise Tracker with Skeletal Motion Detection (https://medium.com/@ahmedrahi/kinect-fitness-playground-creating-an-interactive-exercise-tracker-with-skeletal-motion-detection-ae1423b27c51)
- RISC and CISC: Analyzing Different Approaches to Processor Instructions (https://medium.com/@ahmedrahi/risc-and-cisc-analysing-different-approaches-to-processor-instructions-230a1899d99e)
- The Chiplet Paradigm and What It Means for the Future of Silicon Architectural Design (https://medium.com/@ahmedrahi/the-chiplet-paradigm-and-what-it-means-for-the-future-of-silicon-architectural-design-fa3eea805091)

Languages:
Fluent in English and Urdu; conversational in Punjabi and Arabic.

Certifications: Microsoft Certified: Azure Fundamentals (AZ-900); Microsoft Certified: Azure AI Fundamentals (AI-900); DataCamp AI Engineering; The Leicester Award (Gold), the University of Leicester’s employability and skills award.